Problem Set for 6th March 2025

Exercise 1 Here is the composition rules for linear maps $U \xrightarrow{f} V \xrightarrow{g} W$. Show that

1. if f and g are linear injections, then so is $g \circ f$;
2. if $g \circ f$ is a linear injection, then so is f;
3. if f and g are linear surjections, then so is $g \circ f$;
4. if $g \circ f$ is a linear surjection, then so is g.

Exercise 2 Assume you can prove the following isomorphisms yourself:

$$\begin{aligned}\Phi : \mathrm{Hom}_{\mathbb{F}}(U_1, V) \times \mathrm{Hom}_{\mathbb{F}}(U_2, V) &\to \mathrm{Hom}_{\mathbb{F}}(U_1 \times U_2, V) \\ (f_1, f_2) &\mapsto \begin{bmatrix} U_1 \times U_2 & \to & V \\ (u_1, u_2) & \mapsto & f_1(u_1) + f_2(u_2) \end{bmatrix},\end{aligned}$$

and

$$\begin{aligned}\Psi : \mathrm{Hom}_{\mathbb{F}}(U, V_1) \times \mathrm{Hom}_{\mathbb{F}}(U, V_2) &\to \mathrm{Hom}_{\mathbb{F}}(U, V_1 \times V_2) \\ (g_1, g_2) &\mapsto \begin{bmatrix} U & \to & V_1 \times V_2 \\ u & \mapsto & (g_1(u), g_2(u)) \end{bmatrix}.\end{aligned}$$

Show that

1. $\Phi(f_1, f_2)$ is an injection, only if ($\Rightarrow$) f_1 **and** f_2 are injective;
2. $\Phi(f_1, f_2)$ is a surjection, if ($\Leftarrow$) f_1 **or** f_2 is surjective;
3. $\Psi(g_1, g_2)$ is an injection, if ($\Leftarrow$) g_1 **or** g_2 is injective;
4. $\Psi(g_1, g_2)$ is a surjection, only if ($\Rightarrow$) g_1 **and** g_2 are surjective;
5. **(Optional)** Find counterexamples for converse statements.

Exercise 3 (optional) We know that the composition map

$$\begin{aligned} B : \mathrm{Hom}_{\mathbb{F}}(V, W) \times \mathrm{Hom}_{\mathbb{F}}(U, V) &\to \mathrm{Hom}_{\mathbb{F}}(U, W) \\ (g, f) &\mapsto B(g, f) := g \circ f \end{aligned}$$

is $\mathbb{F}$-bilinear. Show that

1. if g is injective, then $B(g, -)$ is an injection;
2. if f is surjective, then $B(-, f)$ is **still** an injection;
3. if g is surjective and V is coflasque, then $B(g, -)$ is surjective;
4. if f is injective and V is flasque, then $B(-, f)$ is **still** surjective.

> *Here $B(g, -) : f \mapsto B(g, f)$ is a linear map, and so is $B(-, f) : g \mapsto B(g, f)$.*

Definition

- Say Y is flasque, if any linear map from its subspace $Y^{\text{sub}} \to X$ extends to $Y \to X$.
- Say Y is coflasque, if any linear map to its quotient space $X \to Y^{\text{quot}}$ extends to $X \to Y$.

Finite dimensional vector spaces are both flasque and co-flasque.

Exercise 4 (optional) Equivalent definition of being injective or surjection.

1. f is injective, if and only if

$$[f \circ g = f \circ h] \iff [g = h].$$

2. f is surjective, if and only if

$$[g \circ f = h \circ f] \iff [g = h].$$

For **finite dimensional** cases

1. $f : U \to V$ is injective, if and only if there is an $g : V \to U$ such that

$$\left[U \xrightarrow{f} V \xrightarrow{g} U\right] = \left[U \xrightarrow{\mathrm{id}_U} U\right].$$

2. $f : U \to V$ is surjective, if and only if there is an $h : V \to U$ such that

$$\left[V \xrightarrow{h} U \xrightarrow{f} V\right] = \left[V \xrightarrow{\mathrm{id}_V} V\right].$$